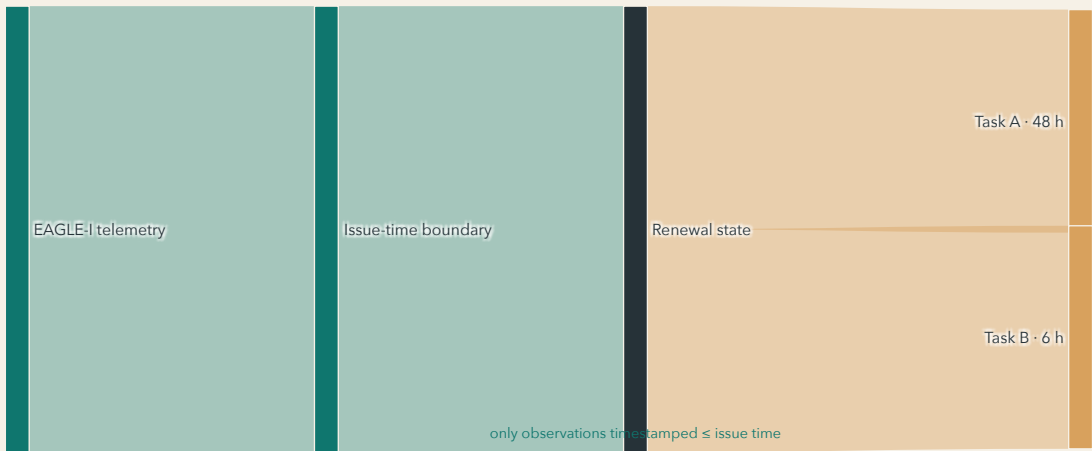


Tideglass

A causal renewal forecaster for county power-outage ratios

A complete Phase-1 system for repeated 48-hour and 6-hour forecasts: sparse county telemetry enters at a declared issue-time boundary, renewal states summarize what is knowable, and two regularized task heads emit continuous outage ratios.

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PROJECT	Tideglass Causal Renewal Forecaster
SUBMISSION DATE	31 August 2026



65,880 combined prediction rows	5 fixed EAGLE-I counties	1 command byte-exact reproduction
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One public signal, five deliberately observable counties

The design starts with the scoring source itself, then treats every absent record as unknown—not as proof of zero outage.

SOURCE AND SPATIAL KEY

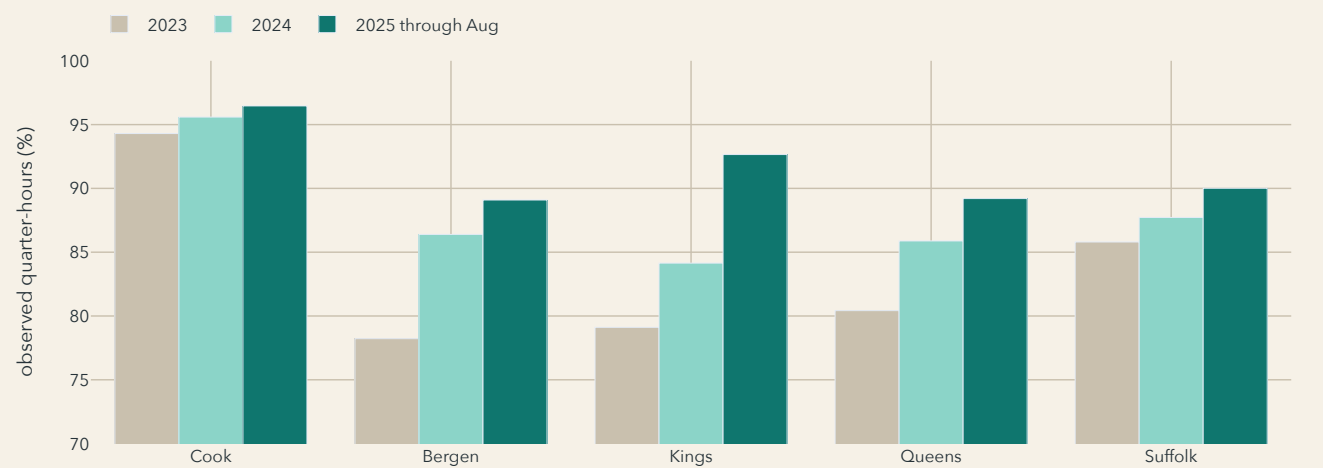
EAGLE-I v4 supplies county outage counts at a nominal 15-minute cadence. `MCC.csv` supplies the organizer-linked fixed customer denominator. We join only on the zero-padded five-digit FIPS code; county and state strings are explanatory fields, never keys.

The chosen set spans one Midwestern metropolitan system and four Northeast systems. Large customer bases reduce ratio quantization; repeated 2023–2025 coverage supports rolling-state construction; the New York / New Jersey cluster also permits a contemporaneous regional context signal.

PREPROCESSING DECISIONS

- Parse raw timestamps as UTC and collapse exact duplicate county/timestamp rows by mean.
- Task A target for validation: mean of available quarter-hour values in the hour ending at the hourly target.
- Task B target: native 15-minute observation.
- Do not densify an absent target to zero; exclude it from local metric denominators.
- At inference, last-observation and lag features use only timestamps at or before the issue time.

FIPS	County	State	MCC customers	Selection role
17031	Cook	Illinois	2,162,007	scale anchor + Midwest contrast
34003	Bergen	New Jersey	418,357	dense Northeast contrast
36047	Kings	New York	915,206	large urban recurrence
36081	Queens	New York	847,158	large urban recurrence
36103	Suffolk	New York	657,677	coastal tail behavior



Observed EAGLE-I rows divided by nominal quarter-hour slots. The rising coverage is why missing rows remain explicit rather than being confused with zero-customer outages.

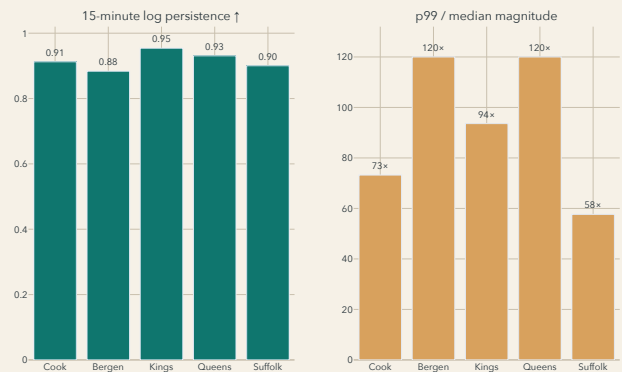
Persistence is strong; magnitudes are not well behaved

The median adjacent-quarter log correlation is 0.913, while each county's 99th percentile is many times its routine median.

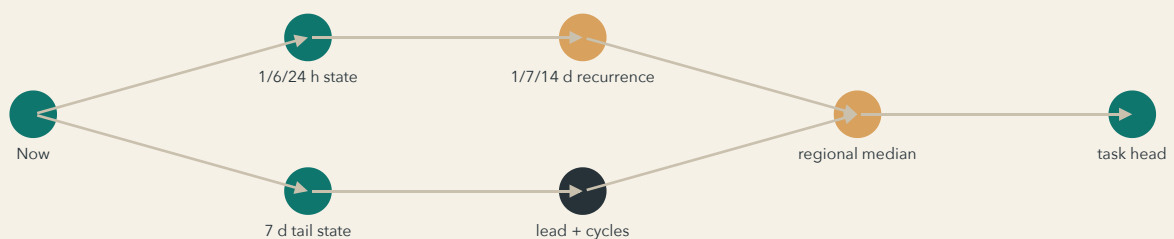
This combination argues against a single unconstrained level regression. Tideglass separates the stable part of the process—county scale, seasonality and recurrence—from an issue-time renewal path that can carry an active event into the short horizon.

FEATURE FAMILIES

- **Current state:** last value, staleness, 1-hour median, 6-hour mean/max/trend.
- **Slow state:** 24-hour median/max and 7-day median / 90th / 99th percentiles.
- **Recurrence:** causal 1-, 7- and 14-day values with missing indicators.
- **Geometry:** lead, square-root/log lead, issue hour, target hour and day-of-year cycles.
- **Spatial context:** the median current ratio across the same five counties.

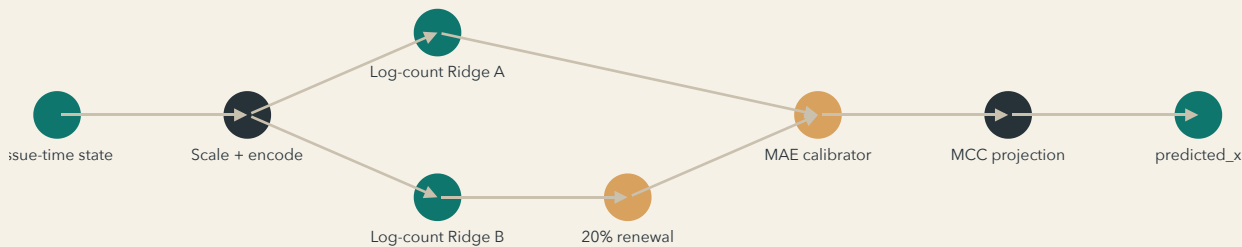


Left: adjacent-quarter persistence on log counts. Right: p99-to-median magnitude ratio (capped visually at 120×). Both are computed before the 2025 test window where applicable.



Why no weather feature? The organizer requires genuine forecasts available at each issue time; target-time observations or stitched hindsight weather would leak. The system uses an auditable outage-renewal path rather than an incomplete archive. The limitation is explicit: an unseen event onset cannot be anticipated until it appears in available telemetry.

Two convex heads, renewal, and historical MAE calibration



TASK A • STRUCTURAL RENEWAL

A task-specific Ridge pipeline predicts `log1p(customers_out)` for +1h through +48h. One-hot county and lead bands combine with standardized continuous features. Long leads naturally lean on slow state, recurrence and calendar structure; no direct persistence override is applied.

TASK B • SHORT-HORIZON RENEWAL

The same regularized convex family is fitted at +15m through +6h. The output is blended with 20% of the last available outage count. That weight was fixed before the sealed release and improves the forward replays' short-horizon tail behavior.

IMBALANCE HANDLING

Outage magnitude is long-tailed rather than cleanly binary. The log target limits domination by rare extremes. Sample weight `1 + min(log1p(y)/10, 1.5)` gives large events up to 2.5× influence while keeping routine intervals central to MAE.

HISTORICAL MAE CALIBRATION

A groupwise weighted-median scale corrects the Ridge heads' positive-event magnitude for MAE: county × lead for Task A, and lead with 500-row shrinkage for Task B. The form was selected on the 2023 forward replay, evaluated once on untouched 2024, then refit from historical 2024 out-of-fold predictions for the sealed model.

PROJECTION

Calibrated counts are divided by the county-specific MCC customer count and clipped to `[0,1]`. The model emits a continuous ratio; it never rounds back to customer lattice points.

206,818

final Task-A training
rows

407,603

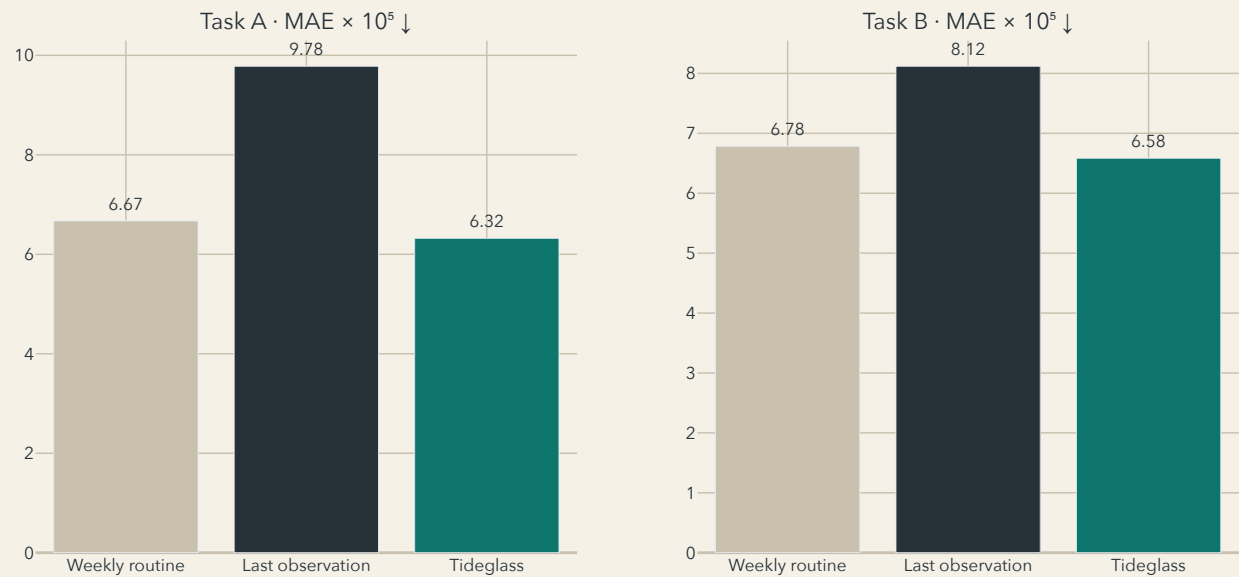
final Task-B training
rows

41,729

fixed seed

Untouched forward replay confirms the calibrated release

The Sep-Nov 2023 expanding-window replay selects the declared calibration form; the untouched Sep-Nov 2024 replay evaluates it. Every replay trains only on earlier information, and missing EAGLE-I targets are excluded rather than invented. These are local public-data diagnostics—not an organizer or leaderboard score.



2024 replay	Task	Rows	MAE	RMSE	Top-decile MAE	Correlation
Weekly routine	A	18,665	0.000067	0.000260	0.000404	0.063
Last observation	A	18,665	0.000098	0.000308	0.000385	0.027
Before calibration	A	18,665	0.000067	0.000256	0.000390	0.141
Tideglass full	A	18,665	0.000063	0.000258	0.000395	0.140
Weekly routine	B	36,658	0.000068	0.000294	0.000430	0.061
Last observation	B	36,658	0.000081	0.000321	0.000372	0.181
Before calibration	B	36,658	0.000067	0.000287	0.000395	0.205
Tideglass full	B	36,658	0.000066	0.000287	0.000404	0.223

Error units are outage-ratio points. Full fold and method records—including bias, p95 absolute error and correlation—are packaged in ``artifacts/validation_metrics.csv``.

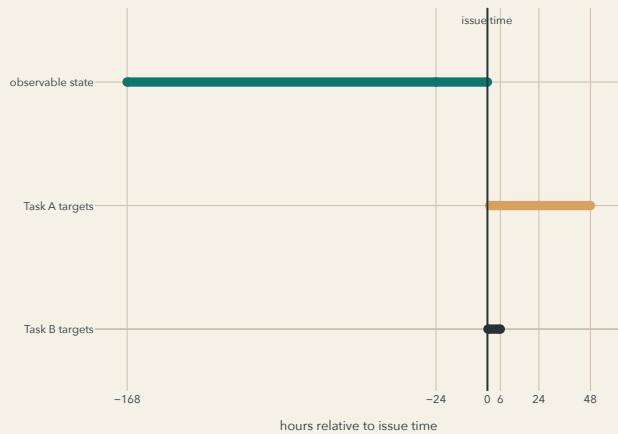
TASK A RESULT

Against last observation, the full model lowers 2024 MAE by 35.4% and RMSE by 16.0%. Calibration further lowers MAE versus the frozen structural head while retaining its lead/calendar signal.

TASK B RESULT

The renewal-plus-calibration path lowers MAE by 18.9% and RMSE by 10.5% versus last observation. Calibration also improves MAE and correlation versus the pre-calibration output, with a modest top-decile MAE tradeoff.

The submitted file is a frozen inference product



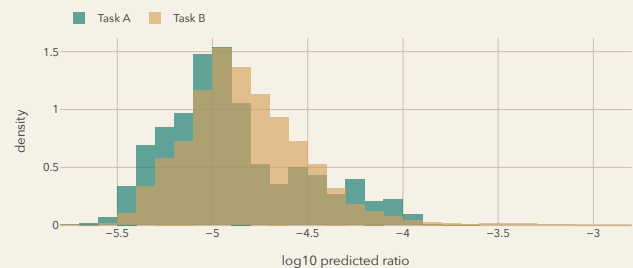
EXACT ROW UNIVERSE

- **Task A:** 92 daily issues × 5 counties × 48 hourly leads = 22,080.
- **Task B:** 365 six-hourly issues × 5 counties × 24 quarter-hour leads = 43,800.
- Primary key: task, FIPS, issue time, target time.
- All timestamps are canonical UTC; every ratio is finite and within bounds.

LEAKAGE CONTROL

Every frozen row stores the latest source timestamp used. An executable assertion verifies ``max_source_timestamp ≤ issue_time``. A target can overlap another batch and remains a separate scored row. The inference file contains features and keys, but no target label.

PREDICTION DISTRIBUTION



Release binding. Model, input and output hashes are validated before packaging. Frozen inference regenerates the submitted CSV byte for byte; retraining is not part of that reproduction path.

predictions.csv SHA-256

ddb7129d0123d5e765a26c78ca0f453b91b749bbd12ceef43edec5b0eb75d1b2

One complete, reproducible release unit

SOURCES AND LICENSES

EAGLE-I / Figshare. "The Environment for Analysis of Geo-Located Energy Information's Recorded Electricity Outages 2014-2025," v4, DOI `10.6084/m9.figshare.24237376.v4`; files for 2023-2025 plus `MCC.csv`; CC BY 4.0.

Organizer specification. Phase-1 submission guideline and illustrative CSV define the task, schema, horizons, five-county choice and test window; specification material only.

Software. NumPy, pandas, scikit-learn, SciPy, Requests, joblib, Plotly, Kaleido, Jinja2 and pypdf. Exact versions and SPDX-style license identifiers are packaged.

KNOWN OPERATING LIMITS

- No unseen weather-onset anticipation.
- Reporting gaps remain missing, so telemetry staleness is a feature.
- MCC denominator revisions require a new bound release.
- Results are county-ratio forecasts, not Phase-2 architecture risk.

EXACT REPRODUCTION

```
cd /path/to/LongCat_Challenge2
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.lock
PYTHONPATH=src python scripts/predict.py \
  --output reproduced_predictions.csv
sha256sum predictions.csv \
  reproduced_predictions.csv
```

FINAL VERIFICATION

- 1 Reproduce `predictions.csv` and confirm the published SHA-256 value.
- 2 Run the release preflight to verify schema, causality, report and archive integrity.
- 3 Confirm the official portal filename, authorization and cutoff.

PDF

technical evidence

CSV

exact predictions

ZIP

code + frozen artifacts

Prepared by Sude Duygun for team LongCat. Method, data provenance, frozen inference, validation evidence and release bindings are complete and included.